

Hotel Bookings Data Analysis

Booking Trends, Cancellation Behaviour and Strategic Recommendations

Dataset: Hotel_bookings_final.csv | Records analysed: 30,000 bookings

1. Introduction and Approach

This report presents an end-to-end analysis of 30,000 hotel booking transactions collected from an online travel platform. The objective was to understand how bookings behave across different sales channels, room categories and property star ratings, to identify the drivers behind cancellations, and to translate these findings into practical recommendations that the business can act on.

The analysis was carried out in Python using pandas for data preparation and aggregation and matplotlib/seaborn for visualisation, with the full workflow run in a Google Colab notebook. A companion dashboard was also built in Looker Studio (Google Data Studio) to allow interactive exploration of the same metrics. The structure of this document follows the three areas the analysis was asked to cover: key observations, root cause analysis, and business recommendations.

1.1 Data Preparation Notes

Before any analysis, the dataset was checked for completeness and consistency. A few points are worth recording here because they shaped how the rest of the analysis was interpreted:

- Out of 30,000 records, 21,672 bookings (72.2%) ended as Confirmed, 6,070 (20.2%) were Cancelled, and 2,258 (7.5%) failed outright (for example, due to a payment issue).
- Check-in and check-out dates were missing for 5,468 records. On closer inspection, these missing values belonged almost entirely to Cancelled and Failed bookings (5,045 and 423 records respectively) — in other words, a stay date was never assigned because the stay never happened. These were left as missing rather than filled in, since filling them would have implied a stay that did not occur.
- The markup applied to the cost price is essentially constant across the dataset, averaging around 30.9% with very little variation. This was an important finding in itself: it means that profit per booking does not vary meaningfully by channel, room type or rating, so any differences in profitability across these groups come from booking volume and conversion rates rather than pricing strategy.
- The customer_id field contains only 499 distinct values across 30,000 bookings (an average of about 60 bookings per customer). This is unusually high for a real customer base and suggests the dataset may be partly synthetic or aggregated at a higher level than individual travellers. This was noted but did not materially affect the channel, room-type or seasonal analysis, which are the focus of this report.

2. Key Observations

2.1 Three Meaningful Trends in the Data

Three patterns stood out as the most consistent and business-relevant findings across the dataset:

Trend 1 — Cancellations spike sharply in four specific months

Cancellation rates are not evenly spread across the year. Four months show cancellation rates well above the typical range of 16-18%: July 2024 (30.3%), August 2024 (28.8%), December 2024 (25.8%) and January 2025 (23.7%). Every other month sits close to the 16-18% baseline. This is not a gradual seasonal drift — it is a sharp spike concentrated in four specific months, which points to something structurally different happening in those periods rather than a smooth seasonal curve.

Trend 2 — Standard rooms are cancelled far more often than Deluxe or Suite rooms, despite similar pricing

Standard rooms are cancelled at a rate of 23.3%, compared with 16.0% for Deluxe and 18.0% for Suite rooms. What makes this notable is that the average booking value across the three room types is almost identical — roughly 24,978 to 25,147 — so price is not the explanation. The gap appears to be about commitment rather than cost: Standard rooms are more likely to be a default or fallback choice, and therefore easier to walk away from.

Trend 3 — Booking channel quality varies more than channel volume

All three channels (Web, Mobile App, Travel Agent) carry a broadly similar number of bookings, but they convert very differently in terms of cancellation, failure and average value — Web converts best, Mobile App fails most often, and Travel Agent cancels most often. The full breakdown is in the table below.

2.2 Booking Patterns Across Channels, Room Types and Star Ratings

By booking channel

Channel	Avg. Booking Value	Cancelled %	Failed %	Confirmed %
Web	28,191	17.6%	5.2%	77.2%
Travel Agent	24,454	27.9%	5.9%	66.2%
Mobile App	21,351	21.6%	10.9%	67.5%

Web is the strongest channel on every dimension that matters — value, cancellation and failure rate. Travel Agent sits in the middle on value but has the worst cancellation rate of the three. Mobile App has the lowest average booking value and by far the highest failure rate, which is the single most visible anomaly in this table.

By room type

Room Type	Avg. Booking Value	Avg. Stay (nights)	Cancelled %
Standard	25,147	~4.0	23.3%
Deluxe	25,005	~4.0	16.0%

Suite	24,978	~4.0	18.0%
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Average booking value and stay length are essentially flat across room types. The only meaningful differentiator is cancellation rate, and Standard rooms are clearly the outlier.

By star rating

Star Rating	Avg. Booking Value	Cancelled %	Failed %
2 Star	24,856	19.8%	7.4%
3 Star	25,210	20.2%	7.2%
4 Star	25,011	20.0%	7.9%
5 Star	25,117	21.3%	7.5%

Star rating has almost no relationship with average booking value — a 2-star property and a 5-star property generate very similar revenue per booking in this dataset. Cancellation rate increases slightly as star rating increases, from 19.8% at 2-star to 21.3% at 5-star, but the effect is modest compared with the channel and room-type differences above.

2.3 Cancellation Behaviour

- Across the full dataset, 20.2% of bookings were cancelled and a further 7.5% failed outright, meaning roughly 28% of all bookings never resulted in a completed stay.
- Cancellation rate does not change meaningfully with how far in advance a booking was made. Whether a booking was made 1 week or up to 2 months before check-in, the cancellation rate stays close to 20% in every lead-time bucket. This rules out 'plans changed over a long booking window' as the main driver of cancellations.
- Of all cancelled bookings, 77.9% (4,731 out of 6,070) were marked as refunded. However, the median refund amount was only about 1.6% of the original booking value, and the average was around 2.4%. In practice this means the platform already operates a largely non-refundable cancellation policy — the 'refund' is closer to a token cashback or fee reversal than a meaningful repayment.
- Coupon usage has almost no effect on cancellation behaviour. Bookings made with a coupon were cancelled at 20.8%, versus 20.1% without a coupon — a difference too small to be meaningful. Average booking value was also very similar with or without a coupon (24,745 vs 25,168).

3. Root Cause Analysis

3.1 What Might Explain the Observed Cancellation Patterns

The cancellation spikes in July, August, December and January do not all share the same underlying cause, and it is worth separating them rather than treating 'peak season' as a single explanation.

- July and August 2024 combine high cancellation rates (30.3% and 28.8%) with above-average booking values (25,502 and 26,065, against an overall average of 25,081). This combination is consistent with speculative summer-holiday booking behaviour — travellers placing more than one booking to secure availability or a price during the busiest planning window, and then cancelling the options they do not use.
- December 2024 and January 2025 also show high cancellation (25.8% and 23.7%), but their average booking values are below the overall average (23,277 and 23,424). This suggests a different mechanism is at work here — most likely cancellations tied to the disruption of holiday-season plans (postponed trips, gift bookings that go unused, or year-end schedule changes) rather than price-driven speculative booking.
- The fact that cancellation rate is flat across lead-time buckets (around 20% regardless of how early the booking was made) reinforces that these spikes are tied to specific calendar periods rather than to how long a customer had to change their mind.
- The very low refund amounts on cancelled bookings (median ~1.6% of value) suggest that financial penalty alone is not currently a strong deterrent — customers appear willing to cancel even though they recover very little of what they paid.
- At the room-type level, Standard rooms cancelling more than Deluxe or Suite (23.3% vs 16.0% / 18.0%) at near-identical price points points to a behavioural rather than financial cause — Standard is more often a 'default' selection that gets revisited or swapped, while Deluxe and Suite bookings tend to reflect a more deliberate choice that the customer is less likely to walk away from.

3.2 Why Some Channels and Property Types Perform Better Than Others

- Web outperforms the other channels on every measure (highest value, lowest cancellation, lowest failure). This is consistent with Web traffic being made up of more deliberate, higher-intent bookings — customers who have already decided what they want and are completing a planned purchase, rather than browsing or comparing.
- Mobile App's failure rate of 10.9% — close to double the 5.2% to 5.9% seen on Web and Travel Agent — is large enough that it is unlikely to be explained by customer intent alone. A failure rate this much higher than the other channels is a strong signal of a technical issue in the mobile booking flow, most likely around payment processing (gateway timeouts, dropped sessions, or a less reliable mobile payment integration).
- Travel Agent's cancellation rate of 27.9% — the highest of the three channels — is consistent with how travel agents typically operate: they often place provisional or exploratory bookings on behalf of clients across multiple properties or dates, intending to confirm only one and cancel the rest once the client decides. This would show up exactly as a higher cancellation rate without necessarily reflecting a problem with the platform itself.

- Star rating showed almost no relationship with booking value, which is somewhat unusual — in most travel platforms, higher-star properties command a price premium. Its absence here suggests that, in this dataset, pricing is driven more by room type, city and seasonal timing than by the star classification of the property.

3.3 Seasonal and Temporal Trends in Booking Value and Stay Length

- Average booking value does vary across the year, but stay length does not. Average stay length is remarkably stable at around 4 nights in every check-in month, ranging only between roughly 3.96 and 4.07 nights. This means that whatever drives the seasonal swing in booking value, it is price per night changing, not how long people stay.
- The highest average booking values in the entire dataset occur in February to April 2025 (roughly 27,300 to 27,800), yet cancellation in these months sits at the normal baseline of 16% to 17.6%. This is an important counterpoint to the July/August pattern: high value on its own does not automatically mean high cancellation. It appears to depend on whether the high value comes from genuine peak pricing on stable bookings (Feb-Apr) or from speculative multi-booking behaviour during a high-demand window (Jul-Aug).
- Put together, the months with the highest cancellation (Jul, Aug, Dec, Jan) correspond to the major holiday and travel-planning periods of the year — summer holidays and the year-end holiday season — while the months with the highest booking values but normal cancellation (Feb-Apr) appear to reflect steady, planned bookings at premium pricing rather than speculative ones.

4. Business Recommendations

4.1 Strategies to Reduce Cancellations

- Season-specific cancellation terms: For Jul, Aug, Dec and Jan check-ins — where cancellation jumps to 24-30% — apply a stricter cancellation window or a partial non-refundable deposit.
- Confirmation step for Travel Agent bookings: This channel has the highest cancellation rate (27.9%), consistent with provisional holds. Require a deposit or explicit confirmation before a booking counts as held.
- Fix the Mobile App checkout flow: Failure rate is 10.9% vs 5.2-5.9% on other channels. A focused review of payment gateway timeouts and session handling could close most of this gap.
- Pre-check-in reminders: Send reminders at 7 and 2 days before check-in to reduce last-minute drop-off.

4.2 Improving Profitability and Increasing Repeat Bookings

- Focus on conversion, not margin: Markup is fixed at ~30.9% across the board, so profitability gains come from reducing the 28% cancel/fail rate, not from pricing changes.
- Redesign coupons for post-stay redemption: Coupons currently show no effect on value or cancellation (20.8% vs 20.1%). Making them redeemable after a completed stay would tie the incentive to follow-through and encourage repeat bookings.
- Use cashback as a loyalty lever: Scale cashback for completed stays and repeat customers, rather than applying it as a flat, unconditional discount.

- Promote Deluxe/Suite over Standard: Same price point, but Standard cancels at 23.3% vs 16.0%/18.0% — shifting demand here improves realised revenue without any pricing change.

4.3 Opportunities to Optimise Pricing, Promotions and Channel Strategy

- Prioritise Web for marketing spend: Highest value (28,191) and best conversion of the three channels — direct acquisition budget here while Mobile App issues are fixed.
- Pair peak pricing with stronger commitment terms: Feb-Apr 2025 proves premium pricing doesn't have to mean high cancellation. Combine Jul/Aug/Dec/Jan peak pricing with prepayment or stricter cancellation terms to retain that value.
- Introduce rating-based pricing tiers: Booking value is flat across 2-5 star properties (24,856-25,210) — an unusual gap that suggests room for a modest premium on 4-5 star properties.
- Small incentive on Standard rooms: A free-upgrade chance or late checkout, offered on Standard bookings (the highest-volume room type), could meaningfully reduce its cancellation rate.

4.4 References

Detailed calculations, full monthly tables, and all underlying charts for the points above are available in the accompanying Python notebook (Google Colab) and the Looker Studio dashboard submitted alongside this report.

4.4 Summary

The dataset points to a platform that is performing reasonably well overall — about 72% of bookings convert into a confirmed stay — but with three areas where targeted action could recover meaningful value: the concentrated cancellation spikes in four specific months, the technical issue affecting Mobile App checkouts, and the higher cancellation rate on Standard rooms and Travel Agent bookings.

The opportunity lies in converting a larger share of the bookings that are already being made — through better-timed cancellation policies, a more reliable mobile experience, and incentives that encourage customers to follow through on the bookings they place.